

Precalculus Exercise Set

Math Dynamics Exploration

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Exercise 1: Function Evaluation

Problem

Evaluate $f(x) = 3x^2 - 4x + 2$ at $x = -1$.

• Solution: $f(-1) = 3(-1)^2 - 4(-1) + 2 = 3 + 4 + 2 = 9$.

Exercise 2: Linear Equation

Problem

Solve $2x + 5 = 13$.

• Solution: $2x = 13 - 5$, $2x = 8$, $x = 4$.

Exercise 3: Quadratic Equation

Problem

Solve $x^2 - 5x + 6 = 0$.

- Solution: Factorize: $(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$.

Exercise 4: Limit

Problem

Find $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

• Solution: Simplify: $\frac{(x-2)(x+2)}{x-2} = x + 2$, so $\lim_{x \rightarrow 2} (x + 2) = 4$.

Exercise 5: Trigonometric Identity

Problem

Prove $\sin^2 \theta + \cos^2 \theta = 1$.

- Solution: By Pythagorean identity, this holds for all θ .

Exercise 6: Inequality

Problem

Solve $x^2 - 4 > 0$.

- Solution: $(x - 2)(x + 2) > 0$, so $x < -2$ or $x > 2$.

Exercise 7: Exponential Function

Problem

Evaluate $2^3 \cdot 2^{-1}$.

• Solution: $2^3 \cdot 2^{-1} = 8 \cdot \frac{1}{2} = 4$.

Exercise 8: Logarithm

Problem

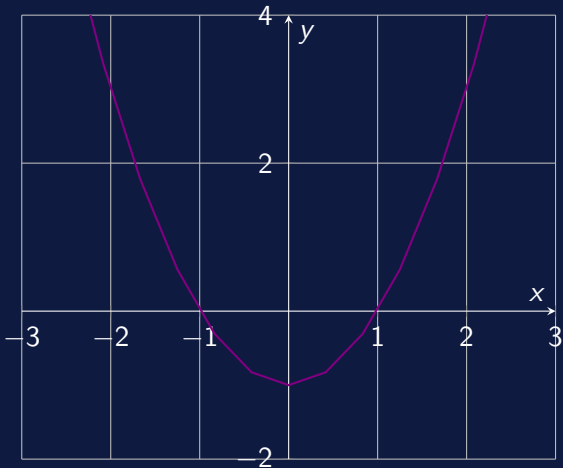
Solve $\log_2(x) = 3$.

• Solution: $x = 2^3 = 8$.

Exercise 9: Graphing

Problem

Sketch $y = x^2 - 1$.



Exercise 10: System of Equations

Problem

$$\text{Solve } \begin{cases} x + y = 5 \\ 2x - y = 1 \end{cases}.$$

• Solution: Add equations: $3x = 6$, $x = 2$. Then $y = 3$.

Conclusion

- These exercises cover key precalculus topics.
- Practice and explore more with Math Dynamics!